

REINFORCEMENT LEARNING ACTIVITY 2

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Part A – RL Problem Setup

1. Choose a game or problem.
Game: Chess

2. Identify the Environment.
Environment: The 8x8 chessboard containing 64 squares, the 32 pieces, and the current state/position of all the pieces on the board.

3. Identify the Agent.
Agent: The AI Chess Player, for example, the AI controlling the White pieces.

4. Identify the Actions.
Actions: Any legal move a piece can make on the board according to the rules of chess, such as moving a Pawn from e2 to e4, or moving a Knight from g1 to f3.

5. Define the Reward Function.

- **Win (Checkmate):** +1
- **Draw (Stalemate):** 0
- **Lose (Checkmated):** -1
- **Optional intermediate rewards:** +0.1 for capturing an opponent's piece, -0.1 for losing a piece.

Part B – Exercise 1.1: Self-Play

1. What do you think would happen when the agent plays against itself?

Initially, the games would look completely random. However, as the agent plays thousands or millions of games against itself, both sides will continuously adapt. Whenever one version of the agent discovers a winning strategy, the other version will eventually learn how to defend against it, pushing both sides to become progressively stronger and smarter.

2. Would the agent learn a different way of playing?

Yes. If an agent plays against a fixed or random opponent, it only learns to exploit that specific opponent's weaknesses. By playing against itself, the agent learns a universally robust way of playing because its opponent is constantly adapting, evolving, and eliminating its own weaknesses.

3. How could self-play affect the agent's strategy?

Self-play removes human bias from the agent's strategy. It forces the agent to develop highly balanced strategies that are strong both offensively and defensively. It might even discover unconventional, "alien-like" strategies that human players have never thought of, similar to what Google's AlphaZero did.

4. What are the possible advantages of self-play?

- **No human data needed:** The agent can learn entirely from scratch without needing a database of human matches.
- **Perfect scaling of difficulty:** The agent is always playing against an opponent of its exact skill level, which is the optimal condition for continuous improvement.
- **Surpassing human limits:** Because it doesn't rely on human examples, it is not constrained by human mistakes or traditional ways of thinking.

5. What problems or limitations might occur during self-play?

- **Overfitting to itself (Mode Collapse):** The agent might develop a very narrow set of strategies that only work well against its own playstyle, making it potentially vulnerable to completely different, random, or unpredictable strategies from a human player.
- **High Computational Cost:** Self-play requires massive amounts of processing power and time, as the agent must simulate millions of games to learn effectively.

- **Slow initial learning:** Because it starts with zero knowledge, the initial phase of learning is essentially random guessing, taking a very long time before any meaningful strategy emerges.